INCH-POUND
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PERFORMANCE SPECIFICATION

LOADER-TRANSPORTER, GUIDED MISSILE: XM501E3 (IMPROVED HAWK)

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of a full-tracked, self-propelled, tactical missile loader for loading and unloading Hawk Guided Missiles onto and from Hawk Missile System Launchers and Pallets.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 1450

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50271 - Plate, Identification, Instruction and Marking, Blank.

DEPARTMENT OF DEFENSE

MIL-H-5440 - Hydraulic Systems, Aircraft Types I and II, Design, Installation, and Data Requirements for.
MIL-H-6083 - Hydraulic Fluid, Petroleum Base, For Preservation and Operation.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility.
MIL-STD-642 - Identification Marking of Administrative, Combat, and Tactical Transport Vehicles.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

MISSILE PURCHASE DESCRIPTION

MPD-7005 - Limit Environmental Criteria for the Anti-Aircraft Guided Missile System-Hawk.

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(Copies of MPD-7005 may be obtained from the U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270.)

DRAWINGS

10879601	- Alignment Checkout Fixture.
11675300	- Loader-Transporter, Guided Missile: XM501E3.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Copies of ANSI Standards may be obtained from the American National Standard Institute, 11 West 42nd Street, 13th Floor, New York, NY 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to the first article inspection in accordance with 4.2.

3.2 Design and construction. Vehicles, components, subassemblies, and assemblies shall be fabricated and assembled into a complete loader in accordance with Drawing 11675300. The weight of the completely assembled loader, serviced with hydraulic fluid, but excluding missiles, fuel, storage items, crane installation, and an operator, shall not exceed 5500 pounds (lbs) [2500 kilograms (kg)] (see 4.5.1).

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3.2.1 Structural interference. All moving joints, fittings, hoses, cables, and structural members shall be free from detrimental interferences (see 4.5.1).

3.2.2 Hydraulic system. The loader hydraulic system shall conform to type II, 3000 pounds per square inch (psi) class of MIL-H-5440 (see 4.5.1).

3.2.2.1 Hydraulic reservoir and lines. The internal cleanliness of the hydraulic reservoir, lines, and fittings shall be maintained throughout the installation of the hydraulic system (see 4.5.1).

3.2.3 Loader electrical system. The loader electrical system shall operate from a regulated 24 volt direct current (dc) supply and shall perform as specified herein under the range of environmental and service conditions to which the loader is exposed during operation. All electrical components shall be securely mounted electrical connections, and the externally exposed connectors shall have watertight seals. The cabling shall show no evidence of broken insulation, broken strands, fraying, or detrimental rubbing against other loader parts (see 4.5.1).

3.2.3.1 Master switch. With the master switch “OFF”, no electrical power shall be provided to operate the loader electrical system. With the master switch “ON”, power shall be supplied to the electrical system and the carrier and superstructural circuits shall operate as specified herein (see 4.5.1).

3.2.3.2 Ignition switch. With the ignition switch at “ON” position, engine starting and operating circuits, engine indicator lights, gages, missile latch and release indicator lights, and fuel pump shall be operable. With the switch at “OFF” position, the operating circuits and fuel pump shall be inoperative (see 4.5.1).

3.2.3.3 Engine starting circuit. With the loader in operating condition, properly serviced, and with the drive selector in “N” position, the starting circuit shall function as follows when the starter switch is operated as follows (see 4.5.1):

a. With the ignition switch “OFF” and the starter switch in the engine starting position, the starter motor shall operate, but the engine shall not start.
b. With the ignition switch at “ON” and the starter switch in the engine starting position, the starter motor shall operate, and the engine shall start.

3.2.3.4 Lights. Normal headlights and taillights, blackout headlights and taillights, and all instrument panel lights shall conform to applicable drawings (see 2.2.2 and 4.5.1).

3.2.3.4.1 Lighting circuits. With light switch assembly on the instrument panel adjusted as shown in table I, lights shall operate as specified herein (see 4.5.1).

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TABLE I. Instrument panel lights.

Switch setting and operating conditions	Light conditions
Main light switch at “off”	No carrier lights operate.
Main light switch at “service drive”	Head and tail lights on.
a. Dimmer switch operated	Headlight beam changes.
b. Brakes applied	Stop light on.
c. Brakes released	Stop light off.
Main light switch at “stop light”	Headlights off, tail lights off.
a. Brakes applied	Stop light on.
b. Brakes released	Stop light off.
Main light switch at blackout (bo) marker	Bo marker and bo tail lights on.
a. Brakes applied	Bo stop light on.
b. Brakes released	Bo stop light off.
Main light switch at bo drive	Bo drive light, bo marker, and bo tail lights on.
a. Brakes applied	Bo stop light on.
b. Brakes released	Bo stop light off.
Main light switch at any setting except “Off” and panel light switch at setting as follows:	Lights operate as specified above, as applicable.
a. “ Off” position	Instrument panel lights off.
b. Park	Instrument panel lights on.
c. Dim	Instrument panel lights dim.
d. Panel bright	Instrument panel lights bright.

3.2.3.4.2 Safety switch. With the safety switch “ON”, and the light switch assembly in normal position, the main light switch (see table I) shall be operable only in the “OFF” or “bo” marker position. The main light switch shall be turnable to “service drive”, or “stop light” positions, as specified in table I, only when the safety switch is simultaneously held in the unlocked position (see 4.5.1).

3.2.3.4.3 Engine indicator circuits. Engine indicator gages and lights shall indicate as shown in table II (see 4.5.1).

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TABLE II. Indicator gages and lights.

Indicator gage or light	Indication
With indication switch "OFF"	
1. Battery generator (batt-gen) indicator 1/	Actual battery voltage.
2. Other gages and indicator lights	Lights off gages do not register.
With ignition switch "ON" - engine not running	
1. Batt-gen indicator	Actual battery voltage.
2. Fuel level gage	Correct quantity of fuel.
3. Engine and transmission (eng and trans) low oil pressure warning lights 1/	On.
4. Transmission, differential, hydraulic, (trans, diff, hyd) and water high temperature (temp) warning lights 1/	Off (unless oil, hydraulic, or water in high temperature range, or unless indicator lights test switch is depressed.)
5. Indicator lights test switch actuated	All indicator warning lights on.
Engine running at fast idle	
1. Batt-gen indicator	Actual generator (voltage regulator) voltage.
2. Fuel level gage	Correct quantity of fuel.
3. All indicator warning lights	Off, except as specified in 4 above.
4. Indicator lights test switch actuated	All indicator warning lights on.

1/ When abbreviations are used to identify indicator lights, they shall be as specified above.

3.2.3.5 Auxiliary power outlet. Auxiliary 24 volt dc power fused at 15 amperes (amps) shall be provided at the utility outlet (see 4.5.1).

3.2.3.6 Engine starting lockout control. With the drive selector lever in any position other than neutral (N), the engine starter electrical circuit shall be inoperative (see 4.5.1).

3.2.4 Superstructure.

3.2.4.1 Superstructure controls. With the superstructure manual operating controls in "NEUTRAL" position, the superstructure shall remain approximately as positioned. Operation of the controls shall provide smooth, positive actuation of the superstructure, throughout all ranges of azimuth, elevation, roll, and extension, under all operational conditions (see 4.5.1).

3.2.4.1.1 Suspension system lockout controls. The suspension lockout control shall be engaged or disengaged with superstructure in transport position. With the suspension lockout control in lockout position, the right and left front and center wheels of the suspension system

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shall be in lockout, and the loader shall either load and unload a pallet, or operate as a crane. With the controls returned to the unlocked position, the suspension system shall return to normal operation (see 4.5.1).

3.2.4.1.2 Crane controls. With the loader engine operating, the speed and direction of cable travel, and swing and elevation of the crane boom shall be controlled by operation of the superstructure hydraulic controls specified in 3.2.4.1. With the roll lockout lever placed in "crane" position, roll control shall be inoperative (see 4.5.1).

3.2.4.2 Superstructure operation. With the superstructure adjusted for normal loader operation, and with the engine running, the superstructure shall respond as specified in table III when the superstructure control levers are moved from their respective neutral positions as specified herein (see 4.5.1).

TABLE III. Normal loader operation.

Lever and lever motion	Superstructure movement	
	Motion	Range of travel
Elevation - Roll		
1. Forward and back	Index boom respectively depresses and elevates	0.7 inch (in.) [18 millimeters (mm)] (minimum) of elevation and depression.
2. Right or left (control centers when released)	Index boom rolls respectively right or left	0.27 in. (6.8 mm) (minimum) in either direction from center level position, except as limited by 3.3.8.2.
Azimuth		
1. Clockwise or counterclockwise (control centers when released)	Main support rotates in azimuth respectively clockwise or counterclock-wise	0.27 in. (6.8 mm) (minimum) in either direction from center position, except as limited by 3.2.4.3.
Extension		
1. Forward or back	Combined link assembly and transfer arms respectively extend and retract	Not applicable.

3.2.4.3 Azimuth and roll control limitations. Azimuth and roll controls shall be operative when the missile is more than 10 in. [25 centimeters (cm)] above the transport position, and shall be inoperative when the missile is less than 2 in. (5 cm) above the transport position (see 4.5.1).

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3.2.4.4 Reduced pressure control. Extension and retraction of the superstructure, during the final phase of extension and the initial phase of retraction, shall be accomplished under reduced hydraulic pressure, not to exceed 630 psi [4.3 Megapascals (MPa)], by energizing the solenoid valve in the hydraulic system extension control loop. The solenoid valve shall energize under conditions "a" and "b" below, de-energize under condition "c", and remain energized under condition "d".

a. Index and transfer arm limit switches are actuated (plungers extended).
b. During extension.
    1. Without missiles, the solenoid valve shall energize at least 0.5 in. (1.3 cm) before the loader hoisting beam forward snap-in latch seats in the upper aft missile pocket, and not before the transfer arm separates from the intermediate link on the limit switch side.
    2. With missiles, the solenoid valve shall energize at least 0.5 in. before the applicable lower aft missile pocket seats against the aft pallet hook, and not before the transfer arm separates from the intermediate link on the limit switch side.
c. During retraction.
    1. Without missiles, the solenoid valve shall de-energize after the loader hoisting beam forward snap-in latch has moved at least 0.5 in. from the upper aft missile pocket, and prior to closure of the gap between the transfer arm and the intermediate link on the limit switch side.
    2. With missiles, the solenoid valve shall de-energize after the lower aft missile pocket has moved at least 0.5 in. from the aft pallet hook, and prior to closure of the gap between the transfer arm and the intermediate link on the limit switch side.
d. During extension and retraction.
    1. With any combination of 1, 2, or 3 missiles, and with the missile(s) latched to the loader simultaneously, the solenoid valve shall remain energized and there shall be no permanent deformation or damage to the loader when the extension control lever is operated in the extended direction (see 4.5.1).

NOTE: Solenoid valves in the roll and elevation systems shall cause the roll and elevation cylinders to become inoperative during the reduced pressure condition.

3.2.4.5 Manual latching of missiles. Positive latching and locking of 1, 2, and 3 missiles to the superstructure hoisting beams shall be accomplished by manually operating applicable latching handles to the latched position. In the latched position, missile latches shall fully extend and remain locked until the rear hoisting beam latches are retracted by manual or electrical release (see 4.5.1).

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3.2.4.6 Manual release of missiles. Latched missiles shall release from the superstructure hoisting beams when the applicable latch release solenoid plunger is manually depressed (retracting aft latch), and when missiles are at rest on the pallet saddles so the weight of the missiles shall be completely off the hoisting beams. In the released condition, aft latches shall be fully retracted (see 4.5.1).

3.2.4.7 Forward missile supports. Outboard forward missile supports shall erect from the stowed position (at rest on loader fenders) to the normal loader position (from support of missiles during loader transport) by rotating the support lever handles from the full forward positions to full back and vice versa (see 4.5.1).

3.2.4.8 Rear center missile support. The rear center missile support and safety bar shall fold down to the stowing position for loader shipment and for crane operation (see 4.5.1).

3.2.5 Superstructure electrical system.

3.2.5.1 Main electrical cable. With the main electrical cable assembly connected between the superstructure and distribution box for normal loader operation, superstructure electrical circuits shall operate normally. With the superstructure cable disconnected and attached to a dummy connector, the superstructure electrical circuits shall not function (see 4.5.1).

3.2.5.2 Superstructure circuits (normal loader operation). With the superstructure adjusted for normal loader operation with missiles, and with the master switch “ON”, the superstructure circuits shall function in accordance with paragraphs 3.2.5.2.1 through 3.2.5.2.3 (see 4.5.1).

3.2.5.2.1 Missile latch and release light circuits. With the ignition switch “OFF”, the “missile-latched” and “missile released” indicator lights shall not illuminate. With the ignition switch “ON”, and with the loader engine and superstructure operating or at rest, the “missile-released” and “missile-latched” lights, indicator-lights-test switch, and dim/bright control shall function in accordance with paragraphs 3.2.5.2.1.1 through 3.2.5.2.1.3 (see 4.5.1).

3.2.5.2.1.1 Missile latch and release lights. The right, center, or left “missile-released” light (red) shall illuminate when the missile latches in the corresponding (right, center, or left) hoisting beam assembly are in the released condition. The right, center, or left “missile-latched” light (green) shall illuminate when the missile latches in the corresponding hoisting beam assembly, are in the latched condition (see 4.5.1).

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3.2.5.2.1.2 Indicator lights test switch. All “missile released” and “missile latched” lights, whether individually “ON” or “OFF”, shall illuminate when the “indicator-lights-test switch” is actuated (switch button held in full depressed position). Indicator lights shall return to their initial condition when the switch button is released (switch returned to normal “OFF” operating position) (see 4.5.1).

3.2.5.2.1.3 Dim bright control. Rotation of dim/bright control from “dim” (maximum counterclockwise position) to “brt” (maximum clockwise position) shall brighten all illuminated “missile-released” and “missile-latched” lights and vice versa. With the control at “dim”, missile-released lights shall be readily visible in the illuminated condition (red) during complete darkness. With the control at “brt”, illumination of “missile-released” lights shall be readily visible in bright sunlight (see 4.5.1).

3.2.5.2.2 Electrical release of missiles. With 1, 2, or 3 missiles latched to the loader, actuation of the “latch release” switch (up position) shall electrically release the one or more selected missiles from the loader when the conditions specified in 3.2.5.2.2.1 are met, and when the full weight of the missiles is off the loader hoisting beam assembly housings (see 4.5.1).

3.2.5.2.2.1 Latch release solenoids. With the loader hoisting beam assemblies in the “missile-latched” condition as specified herein, energizing the “latch release” solenoids shall retract the latch locks so that the spring-loaded latching assemblies will return to the “missile released” condition when the weight of the missiles is not suspended from the latches. Solenoids shall energize only when the following conditions are met (see 3.2.5.2.2 and 4.5.1):

a. Superstructure master switch is actuated to “ON”.
b. Index and transfer arm limit switches are actuated to “ON” (plunger extended) (see 3.2.5.2.3).
c. Applicable circuit breakers are on (toggle “up” position).
d. Missile “latch release” switch is actuated to “ON”.

3.2.5.2.3 Index boom and transfer arm limit switches. The index boom-intermediate link limit switch shall actuate only when the superstructure is extended until the link assembly is raised from the index boom trunnion beyond a point where the limit-switch plunger is extended. The transfer arm-intermediate link limit switch shall actuate only when the transfer arm is separated from the intermediate link assembly beyond a point where the limit-switch plunger is extended (see 4.5.1).

3.2.6 Stowed material. Unless otherwise specified (see 6.2), all on vehicle material (OVM) shall be stowed on the vehicle in the space provided to assure that it will not interfere with the operation of the vehicle and/or components in any manner. For stowing purposes, the manufacturer installed OVM, as furnished by the vehicle contractor, and the depot installed OVM, as furnished by the Government, shall be of the latest configuration and production (see 4.5.1).

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3.2.7 Surface treatment. Unless otherwise specified (see 6.2), exterior surfaces of the loader shall be cleaned, treated, primed, and painted in accordance with the manufacturer's standard practice. The color required shall be as specified in the contract or order (see 6.2). When specified (see 6.2), CARC paint shall be required (see 4.5.1).

3.2.8 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-642 (see 4.5.1).

3.2.9 Name, shipping, and identification data plates. The information to be included shall be in accordance with applicable drawings or as specified by the procuring activity. The plates shall meet the performance requirements of A-A-50271 (see 4.5.1).

3.2.10 Fire extinguisher. When mounted in accordance with the drawings, the extinguisher shall be readily accessible and shall operate without malfunction (see 4.5.1).

3.2.11 Workmanship. The loader shall be free from such defects as breaks, bends, cracks, deep cuts, rust, chipped or spotty paint, damaged fastenings, looseness of attaching parts, misalignment, and binding or detrimental interference of parts. All bolted joints shall have reasonably flat mating surfaces that are free from burrs, weld spatters, and other significant surface irregularities (see 4.5.1).

3.3 Performance. The loader shall operate as specified herein without special equipment, or any evidence of excessive vibration, overheating of the engine, powertrain, or assemblies. There shall be no evidence of leaking of exhaust, lubricants, water, fuel, or hydraulic fluids, looseness, misalignment, misadjustment, binding of parts, or any other defect or malfunction during operation (see 4.5.3).

3.3.1 Pre-test break-in run (all vehicles). After being subjected to the pre-test break-in run specified in 4.2.3, there shall be no excessive grinding or backlash in the gear trains of the loader, and all suspension wheel bearings shall be correct or readjusted as required. There shall be no indication of damaged components or maladjustments that may cause faulty vehicle operation.

3.3.2 Fuel system. The fuel system shall maintain a fuel supply to the engine when tested in accordance with 4.5.3.1.

3.3.3 Power package and power train. When tested in accordance with 4.5.3.2, the power package shall function through all gear and speed ranges without loss of lubricants or damage that may cause failure of the loader. The power train and associated controls shall operate without binding of linkages, loss of lubricants, grabbing, and chattering, or slippage when controls are applied to control, stop, and hold the vehicle.

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3.3.4 Hydraulic pressure.

3.3.4.1 Start-up hydraulic pressure. When tested in accordance with 4.5.3.3.1, the hydraulic pressure registered on the loader hydraulic oil pressure gauge shall not exceed 50 psi [340 kiloPascals (kPa)], when the hydraulic load is removed from the engine.

3.3.4.2 Operating hydraulic pressure. When tested in accordance with 4.5.3.3.2, the operating pressure of the engine driven hydraulic pump shall be 3000 (+200, -100) pounds per square inch gage (psig) (21 +1.4, -0.7 MPa). The maximum system pressure shall be limited to 3400 psig (23 MPa) by a system relief valve.

3.3.4.3 Proof pressure. The hydraulic system shall withstand the proof pressures specified in 4.5.3.3.3 without damage.

3.3.5 Radio interference suppression. The complete loader shall be suppressed to prevent interference to high-sensitivity radio communication equipment in the frequency range of 0.15 megacycles (mc) to 500 mc, in accordance with the applicable provisions of MIL-STD-461 for tactical vehicles (see 4.5.3.4).

3.3.6 Environmental conditions. The loader shall perform normally when subjected to the environmental conditions specified in Missile Purchase Description MPD-7005 (see 4.5.3.5).

3.3.7 Loader mobility.

3.3.7.1 Speed. When tested in accordance with 4.5.3.6.1, the loader shall achieve a top speed of not less than 10 miles per hour (mph) [16 kilometers per hour (km/h)] with the gear selector in drive (D) position, and not less than 5 mph (8 km/h) with the gear selector in hilly (H) position. The loader shall perform without malfunction or transmission shifting.

3.3.7.2 Creeping. When tested in accordance with 4.5.3.6.2, the loader shall be capable of smooth creeping at speeds not to exceed 6 in. (15 cm) per second.

3.3.7.3 Acceleration. The loader shall be capable of accelerating from a standing start, with engine at idle, to 10 mph (16 km/h) in not more than 45 seconds (see 4.5.3.6.3).

3.3.7.4 Turning. The loader shall be capable of turning as specified in 4.5.3.6.4, with the carrier remaining within a 40 foot (ft) [12 meter (m)] diameter circle.

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3.3.7.5 Braking.

3.3.7.5.1 Brake controls. With the brake control levers fully released or locked, the tops of the brake lever handles shall be in alignment with each other within 0.5 in. (1.3 cm). The force required to lock the brakes shall not vary between the two levers by more than 5 pounds force (lbf) [22 Newtons (N)]. With each brake lever in fully locked position, there shall be at least 2 notches of travel remaining (see 4.5.3.6.5.1).

3.3.7.5.2 Brake lever turning force. When tested in accordance with 4.5.3.6.5.2, the brake lever force necessary to start a turn at all temperatures down to 32 degrees Fahrenheit (°F) [0 degrees Celsius (°C)], shall not be more than 40 lbf (180 N), and the lever force necessary to hold the turn shall not be more than 30 lbf (130 N). The increase in lever force at all temperatures between 32°F (0°C) and -65°F (-54°C) shall not exceed 20 percent.

3.3.7.5.3 Stopping distance. When tested in accordance with 4.5.3.6.5.3, the loader shall stop within 10 ft (3 m) from the point of brake application.

3.3.7.5.4 Holding. When tested in accordance with 4.5.3.6.5.4, the service brakes shall hold the vehicle stationary for not less than 3 minutes for both up and down grades. The loader shall remain stationary for 3 minutes with only the parking brake applied for up and down grades. The loader shall remain stationary when both brakes are applied and the engine is accelerated to 1000 revolutions per minute (rpm).

3.3.7.6 Drift. When tested in accordance with 4.5.3.6.6, drift shall not exceed 3 ft (0.9 m) in either direction.

3.3.7.7 Environmental obstacles. The loader shall negotiate the environmental obstacles specified in 4.5.3.6.7.1 through 4.5.3.6.7.8, and shall operate normally, without malfunction, or injury to the operator.

3.3.7.8 Towing. When tested in accordance with 4.5.3.6.8, the loader shall achieve a minimum speed of not less than 2 mph (3 km/h).

3.3.8 Superstructure testing.

3.3.8.1 Superstructure alignment. When tested in accordance with 4.5.3.7.1, the loader hoisting beam latches shall be in alignment with the applicable missile latching points. The latches shall engage the fixture pockets of the alignment checkout fixture readily and without need of external force to the hoisting beams. When the manual latch release is operated, the latches shall release readily without binding.

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3.3.8.2 Superstructure settling. When subjected to the conditions specified in 4.5.3.7.2, the superstructure shall not settle more than 1 in. (2.5 cm) (measured vertically at missiles) in one minute.

3.3.8.3 Superstructure operating speeds. When subjected to the conditions specified in 4.5.3.7.3, the superstructure shall perform as specified in table IV.

TABLE IV. Superstructure operating speeds.

Operation	Engine speed (rpm)	Traverse time (entire range) seconds
Elevation cylinder		
1. Elevate	2000 - 2500	10 to 21
2. Depress	2000 - 2500	10 to 21
Roll cylinder		
1. Full right to full left	2000 - 2500	15 to 21
2. Full left to full right	2000 - 2500	15 to 21
Azimuth cylinder		
1. Rotate from full right to full left	1500 - 2000	4 to 12
2. Rotate from full left to full right	1500 - 2000	4 to 12
Extension cylinder		
1. Extend (loader indexed to pallet on ground)	3000 - 3200	15 to 30 1/
2. Retract (loader indexed to pallet on ground)	3000 - 3200	15 to 35 1/

1/ Entire range includes reduced pressure travel of missiles on pallet.

3.3.9 Loading operations.

3.3.9.1 Missile handling. The loader shall be capable of indexing to the launcher and loading or unloading the launcher with 1, 2, or 3 missiles under the conditions specified in 4.5.3.8.1.

3.3.9.2 Pallet loading. The loader shall be capable of loading and unloading a pallet under the conditions specified in 4.5.3.8.2.

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3.3.9.3 Proof load. The loader shall be capable of withstanding loads imposed by 110 percent of the normal improved missile weight (1526 to 1580 lbs (692 to 717 kg)) (see 4.5.3.8.3).

3.3.10 Crane operation. When subjected to the tests of 4.5.3.9 through 4.5.3.9.5, the crane shall operate normally without malfunction. All moving joints, fittings, hoses, electrical cables, stays, and structural members shall be free from binding or detrimental interference during operation.

3.3.10.1 Hoisting cable. When tested in accordance with 4.5.3.9.1, the hoisting cable shall raise and lower the maximum loads.

3.3.10.2 Boom swing. When tested in accordance with 4.5.3.9.2, the crane boom (ramp beam) shall swing clockwise and counterclockwise, respectively.

3.3.10.3 Boom elevation. When tested in accordance with 4.5.3.9.3, the crane boom shall elevate and depress, respectively.

3.3.10.4 Hoisting hook position. When tested in accordance with 4.5.3.9.4, the lowest point of the hoisting hook shall not be higher than 12 in. (30 cm) from the ground with the hoisting cable fully extended. With the hoisting cable fully retracted, the hoisting hook shall not be lower than 150 in. (380 cm) from the ground.

3.3.10.5 Rated crane capacity. When tested in accordance with 4.5.3.9.5, the loader shall maintain stability at all times when lifting the specified loads. There shall be no permanent deformation or damage to the loader.

3.3.11 Auxiliary equipment. The loader shall be compatible with the equipment specified in 6.3 (see 4.5.3.10).

4. VERIFICATION

4.1 Classification of inspections. The inspection conditions specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspections (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on preproduction or initial production samples as specified when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.4 (see table V) and the tests of 4.2.3 through 4.5.3.9.5 (see table VI).

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TABLE V. Classification of defects.

Category	Defect	Method of examination
Major:		
101	Design and construction not as specified (see 3.2).	Visual and SIE 1/
102	Evidence of structural interference (see 3.2.1)	Visual
103	Hydraulic system not as specified (see 3.3.2 and 3.2.2.1).	Visual and functional
104	Electrical system not as specified (see 3.2.3).	Visual and functional
105	Master switch not as specified (see 3.2.3.1).	Visual and functional
106	Ignition switch not as specified (see 3.2.3.2).	Visual and functional
107	Engine starting circuit not as specified (see 3.2.3.3).	Visual and functional
108	Lights not as specified (see 3.2.3.4).	Visual and functional
109	Lighting circuits not as specified (see 3.2.3.4.1).	Visual and functional
110	Safety switch not as specified (see 3.2.3.4.2).	Visual and functional
111	Engine indicator circuits not as specified (see 3.2.3.4.3).	Visual and functional
112	Auxiliary power outlet missing or not as specified (see 3.2.3.5).	Visual and functional
113	Engine starts with drive selector in position other than neutral (see 3.2.3.6).	Visual and functional
114	Superstructure controls not as specified (see 3.2.4.1).	Visual and functional
115	Suspension system lockout controls not as specified (see 3.2.4.1.1).	Visual and functional
116	Crane controls not as specified (see 3.2.4.1.2).	Visual and functional
117	Superstructure operation not as specified (see 3.2.4.2).	Visual, functional and SIE

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TABLE V. Classification of defects - Continued.

Category	Defect	Method of examination
118	Azimuth and roll control limitations not as specified (see 3.2.4.3).	Visual, functional and SIE
119	Reduced pressure control for superstructure not as specified (see 3.2.4.4).	Visual, functional and SIE
120	Manual latches not as specified (see 3.2.4.5).	Visual and functional
121	Latch release not as specified (see 3.2.4.6).	Visual and functional
122	Forward missile supports not as specified (see 3.2.4.7).	Visual and functional
123	Rear center missile support does not fold down into stowing position (see 3.2.4.8).	Visual and functional
124	Electrical cable not as specified (see 3.2.5.1).	Visual and functional
125	Superstructure circuits not as specified (see 3.2.5.2).	Visual and functional
126	Missile latch and release light circuits not as specified (see 3.2.5.2.1).	Visual and functional
127	Missile latch and release lights not as specified (see 3.2.5.2.1.1).	Visual and functional
128	Indicator lights test switch not as specified (see 3.2.5.2.1.2).	Visual and functional
129	Dim/bright control not as specified (see 3.2.5.2.1.3).	Visual and functional
130	Missiles do not release when specified (see 3.2.5.2.2).	Visual and functional
131	Latch release solenoids not as specified (see 3.2.5.2.2.1).	Visual and functional
132	Limit switches not as specified (see 3.2.5.2.3).	Visual and functional
133	Stowed material not as specified (see 3.2.6).	Visual

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TABLE V. Classification of defects - Continued.

Category	Defect	Method of examination
134	Surface treatment not as specified (see 3.2.7).	Visual
135	Marking mission or not as specified (see 3.2.8).	Visual
136	Data plates mission or not as specified (see 3.2.9).	Visual
137	Fire extinguisher missing or not as specified (see 3.2.10).	Visual
Minor: 201	Workmanship not as specified (see 3.2.11).	Visual

1/ SIE = Standard Inspection Equipment.

TABLE VI. Inspections.

Title	Requirement	Inspection	First article	Conform-ance
Pre-test break-in run	3.3.1	4.2.3	X	X
Performance, general	3.3	4.5.3	X	X
Fuel system	3.3.2	4.5.3.1	X	X
Power package and power train	3.3.3	4.5.3.2	X	X
Start-up hydraulic pressure	3.3.4.1	4.5.3.3.1	X	X
Operating hydraulic pressure	3.3.4.2	4.5.3.3.2	X	X
Proof pressure	3.3.4.3	4.5.3.3.3	X	X
Radio interference suppression	3.3.5	4.5.3.4	X	
Environmental conditions	3.3.6	4.5.3.5	X	
Speed	3.3.7.1	4.5.3.6.1	X	X
Creeping	3.3.7.2	4.5.3.6.2	X	X
Acceleration	3.3.7.3	4.5.3.6.3	X	X
Turning	3.3.7.4	4.5.3.6.4	X	X
Brake controls	3.3.7.5.1	4.5.3.6.5.1	X	X
Brake lever turning force	3.3.7.5.2	4.5.3.6.5.2	X	
Stopping distance	3.3.7.5.3	4.5.3.6.5.3	X	X
Holding	3.3.7.5.4	4.5.3.6.5.4	X	X
Drift	3.3.7.6	4.5.3.6.6	X	X
Trench	3.3.7.7	4.5.3.6.7.1	X	
Vertical obstacle	3.3.7.7	4.5.3.6.7.2	X	
Blast hole	3.3.7.7	4.5.3.6.7.3	X	
Rough terrain	3.3.7.7	4.5.3.6.7.4	X	
Snow	3.3.7.7	4.5.3.6.7.5	X	
Water	3.3.7.7	4.5.3.6.7.6	X	

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TABLE VI. Inspections - Continued.

Title	Requirement	Inspection	First article	Conform-ance
Slopes	3.3.7.7	4.5.3.6.7.7	X	X
Side slopes	3.3.7.7	4.5.3.6.7.8	X	
Towing	3.3.7.8	4.5.3.6.8	X	
Superstructure alignment	3.3.8.1	4.5.3.7.1	X	X
Superstructure settling	3.3.8.2	4.5.3.7.2	X	
Superstructure operating speeds	3.3.8.3	4.5.3.7.3	X	
Missile handling	3.3.9.1	4.5.3.8.1	X	X
Pallet loading	3.3.9.2	4.5.3.8.2	X	X
Proof load	3.3.9.3	4.5.3.8.3	X	X
Crane operation	3.3.10	4.5.3.9	X	
Hoisting cable	3.3.10.1	4.5.3.9.1	X	
Boom swing	3.3.10.2	4.5.3.9.2	X	
Boom elevation	3.3.10.3	4.5.3.9.3	X	
Hoisting hook position	3.3.10.4	4.5.3.9.4	X	
Rated crane capacity	3.3.10.5	4.5.3.9.5	X	
Auxiliary equipment	3.3.11	4.5.3.10	X	

4.2.1 Preproduction inspection. When specified (see 6.2), one preproduction vehicle shall be inspected for conformance to all requirements contained in this specification. The inspection shall be performed at a location approved by the Government. Failure of the preproduction vehicle to comply with any of the requirements specified herein shall be cause for refusal by the Government to conduct a retest until corrective action taken by the contractor has been approved by the Government.

4.2.2 Initial production inspection. Unless otherwise specified (see 6.2), initial production vehicles shall be subjected to a special process inspection, conducted by the Government representatives during fabrication of the first production vehicle, to determine conformance of materials and workmanship with drawing and specification requirements. After completion of the first vehicle inspection, a second vehicle selected from the first months production, or one of the first 10 vehicles produced, shall be subjected by the Government to examinations and tests specified in table V and table VI. Subsequent to examinations and tests, the selected vehicle shall be subjected by the Government to the 250 hours test specified in table VII at a site approved by the Government, which will require a maximum of 60 days. Delays caused by vehicle breakdown due to poor quality of workmanship or material, or failure of the contractor to comply with specifications or drawing requirements, shall not be the basis for adjustment of the contract performance dates, delivery schedule, or the contract price. Failure of the vehicle to comply with any of the requirements specified or any deficiency of a workmanship

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or materials nature found on the vehicle during or as a result of the 250 hour test shall be considered prima facie evidence that all vehicles already accepted prior to completion of the 250 hour test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies shall be corrected by the contractor at no cost to the Government.

TABLE VII. 250 hour test.

Cycle breakdown for loader operation	
1. Missile transfer	70 hours
2. Missile transport	70 hours
3. Miscellaneous	60 hours
Cycle breakdown for mobile crane	
1. Load handling	12 hours
2. Maneuvering	12 hours
3. Miscellaneous	12 hours
Cycle breakdown for use as utility vehicle	
1. Maneuvering	6 hours
2. Hauling extremely heavy load	3 hours
3. Miscellaneous	5 hours

4.2.3 Pre-test break-in run (all vehicles). Unless otherwise specified (see 6.2), the loader shall be subjected to a pre-test run. Prior to pre-test run, the loader shall be completely assembled and all adjustments completed. The loader, including all components requiring lubrication, shall be lubricated and serviced as required, using standard military products as shown on the lubrication drawings. Prior to start of pre-test run, or any time during the run when the engine has been stopped for a period of 15 minutes, the engine, when started, shall be warmed-up for at least 3 minutes, and all engine indicator lights and gages shall indicate normal operating conditions. Each loader, without missiles and with the superstructure in normal transport position, shall be given a pre-test run on smooth, level, hard-surfaced roads. The loader shall be operated as specified in table VIII. Before operating in reverse (R), each time during the pre-test run, the loader shall be brought to a stop, placed in neutral and the engine allowed to idle for not less than two minutes. Nonconformance to 3.3.1 shall constitute failure of the pre-test run.

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TABLE VIII. Speeds and distance for pre-test run.

Division of run	Engine rpm	Driver-selector position	Distance yards	(meters)
a	1500	H	880	(800)
b	2200	H	880	(800)
c	2000	R	17	(15)
d	1500	D	880	(800)
e	2400	D	880	(800)
f	2000	R	17	(15)

NOTE: H - Hilly, D - Drive, R - Reverse.

4.3 Conformance inspection. The conformance inspection shall include the examinations of 4.4, and the tests of 4.2.3, 4.5.3 thru 4.5.3.3.3, 4.5.3.6.1 thru 4.5.3.6.5.1, 4.5.3.6.5.3 thru 4.5.3.6.6, 4.5.3.6.7.7, 4.5.3.7.1, and 4.5.3.8.1 thru 4.5.3.8.3 (see table VI).

4.4 Examination.

4.4.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI/ASQC Z1.4. Any redesign or modification of the contractor's standard to comply with specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements of requirements of 3.2 through 3.2.13 as listed in table V. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Method of inspection.

4.5.1 Defects. Conformance to 3.2 through 3.2.13, shall be determined by examination for the defects listed in table V. Examination shall be visual, tactile, functional, or by measurement with Standard Inspection Equipment (SIE).

4.5.2 Test conditions. Prior to tests, the complete loader shall be serviced with specified fuel, lubricants, grease, and fluids. Unless otherwise specified, the loader shall be tested on relatively smooth, level, hard surfaced roads.

4.5.3 Performance. The loader shall be subjected to all of the tests specified in this section (see 3.3).

4.5.3.1 Fuel system. To determine conformance to 3.3.2, the loader shall ascend and descend 60 percent grade slopes in forward and reverse gear.

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4.5.3.2 Power package and power train test. To determine conformance to 3.3.3, the loader shall be operated through all ranges of operation and all controls shall be utilized.

4.5.3.3 Hydraulic pressure.

4.5.3.3.1 Start-up hydraulic pressure. To determine conformance to 3.3.4.1, the hydraulic pressure gage shall be checked with the starter switch in engine starting position and with the engine turning over.

4.5.3.3.2 Operating hydraulic pressure. To determine conformance to 3.3.4.2, the hydraulic pump, serviced with hydraulic fluid conforming to MIL-H-6083, shall be operated for not less than 15 minutes and checked for operating pressures using the loader hydraulic pressure gage.

4.5.3.3.3 Proof pressure. To determine conformance to 3.3.4.3, the hydraulic pump shall be operated for not less than 15 minutes and checked for relief and proof pressures using an external pressure source and gages. Burst pressure shall not be applied.

4.5.3.4 Radio interference suppression. To determine conformance to 3.3.5, the loader shall be tested for radio interference in accordance with the applicable provisions of MIL-STD-461 for tactical vehicles.

4.5.3.5 Environmental conditions. To determine conformance to 3.3.6, the loader shall be tested in accordance with the applicable provisions of Missile Purchase Description MPD-7005.

4.5.3.6 Loader mobility.

4.5.3.6.1 Speed. To determine conformance to 3.3.7.1, the loader, with the gear selector in drive (D) position, shall start out with the transmission in low range, then shift to transmission intermediate range, then shift to transmission high range, after which a maximum speed shall be achieved and maintained for 15 minutes. The gear selector shall then be set to hilly (H) position, after which a maximum speed shall be achieved and maintained for 15 minutes.

4.5.3.6.2 Creeping. To determine conformance to 3.3.7.2, the gear selector shall be set to drive (D) position and the loader shall be operated at a speed not greater than 6 in. (15 cm) per second. The brakes may be utilized as required. The loader shall be checked for smoothness of movement during operation.

4.5.3.6.3 Acceleration. To determine conformance to 3.3.7.3, the loader shall be accelerated, from a standing start with the engine idle, at the most rapid speed possible and the rate of speed measured.

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4.5.3.6.4 Turning. To determine conformance to 3.3.7.4, the loader shall be positioned in the center of a 40 ft (12 m) diameter circle, after which, in forward or reverse direction, the loader shall execute full 360 degree turns to the left and right directions.

4.5.3.6.5 Braking.

4.5.3.6.5.1 Brake controls. The brake control levers shall be fully released or locked and the difference in alignment between the tops of the brake levers measured. The force required to lock each of the two brake levers shall be measured. With each brake lever in fully locked position, the number of travel time notches shall be determined. Nonconformance to 3.3.7.5.1 shall constitute failure of the test.

4.5.3.6.5.2 Brake lever turning force. To determine conformance to 3.3.7.5.2, the brake lever force necessary to start and hold a turn shall be measured with the loader operating in a temperature of 32°F (0°C). This procedure shall then be repeated at a temperature of -65°F (-54°C).

4.5.3.6.5.3 Stopping distance. To determine conformance to 3.3.7.5.3, the loader shall be operated at a speed of 10 mph (16 km/h), after which the brakes shall be applied and the distance from the point of brake application measured.

4.5.3.6.5.4 Holding. The loader shall be loaded with the maximum load or a simulated load of equal size and weight, and on a 40 percent grade longitudinal slope, the service brake shall be applied for 3 minutes for the loader headed both up and down grade. The brakes shall then be locked for parking for 3 minutes (while on the slope), and all other holding devices shall be made inoperative, for up and down grades. The loader shall then be moved to a smooth, level, hard surface, and the engine shall be reved to a minimum of 1000 rpm, with the gear selector in hilly (H) position, and with both brakes fully applied. Nonconformance to 3.3.7.5.4 shall constitute failure of the test.

4.5.3.6.6. Drift. To determine conformance to 3.3.7.6, the loader, while on a smooth, level, hard surface, shall be operated straight forward and straight backward (reverse gear) 100 ft (30 m) in both directions. All brakes shall be released and the steering levers shall be in the full forward position. The loader operating speed shall fall between 5 and 10 mph (8 and 16 km/h) for the full 100 ft distance in both directions.

4.5.3.6.7 Environmental obstacles.

4.5.3.6.7.1 Trench. The loader shall cross a 25 in. (64 cm) wide trench at reduced speed. Nonconformance to 3.3.7.7 shall constitute failure of the test.

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4.5.3.6.7.2 Vertical obstacle. The loader shall be operated over a 10 in. (25 cm) high vertical obstacle at reduced speed. Nonconformance to 3.3.7.7 shall constitute failure of the test.

4.5.3.6.7.3 Blast hole. The loader shall be operated across a generally circular blast hole, 5 ft (1.5 m) in diameter and 18 in. (46 cm) deep, at reduced speed. Nonconformance to 3.3.7.7 shall constitute failure of the test.

4.5.3.6.7.4 Rough terrain. The loader shall be operated across hard, boulder strewn terrain, with obstacles up to 5 in. (13 cm) in height, at a speed of 5 mph (8 km/h). Nonconformance to 3.3.7.7 shall constitute failure of the test.

4.5.3.6.7.5 Snow. The loader shall meet all of the performance requirements specified herein when operated in wet snow, 12 in. (30 cm) or more in depth, for 8 hours. Nonconformance to 3.3.7.7 shall constitute failure of the test.

4.5.3.6.7.6 Water. The loader shall ford a salt or fresh water crossing, 20 in. (50 cm) or more in depth (including sinkage and wave height). Nonconformance to 3.3.7.7 shall constitute failure of the test.

4.5.3.6.7.7 Slopes. The loader, weighted with missiles or a simulated maximum load, in both forward and reverse gear, shall be operated up a relatively smooth, longitudinal 40 percent grade. The missiles shall then be removed from the loader, and the loader, in forward and reverse gear, shall be operated up a relatively smooth, longitudinal 60 percent grade from a standing start on the grade. Nonconformance to 3.3.7.7 shall constitute failure of the test.

4.5.3.6.7.8 Side slope. The loader, weighted with 3 missiles, shall be operated at maximum speed across a smooth, 30 percent grade side slope. One missile shall then be removed so that the load is unbalanced in the downhill direction, after which the loader shall be operated at maximum speed across a smooth, 18 percent grade side slope. Nonconformance to 3.3.7.7 shall constitute failure of the test.

4.5.3.6.8 Towing. The loader shall be chained to a load, for which the breakaway drawbar pull is not less than 5000 lbs (2300 kg) and the continuous drawbar pull is not less than 2500 lbs (1130 kg), and operated for a distance of 1000 ft (300 m) over relatively level, smooth, firm terrain. The loader speed shall be observed to determine conformance to 3.3.7.8.

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4.5.3.7 Superstructure testing.

4.5.3.7.1 Superstructure alignment. The empty loader shall be indexed to the 10879601 alignment checkout fixture. The superstructure shall be extended until the hoisting beams are in latching position. The hoisting beam latch levers shall then be operated until the latches are properly engaged in the fixture pockets. Nonconformance to 3.3.8.1 shall constitute failure of the test.

4.5.3.7.2 Superstructure settling. To determine conformance to 3.3.8.2, the loader shall carry 3 missiles, the elevation and extension cylinder may be extended to any positions within their respective ranges, the elevation and extension valves shall be in neutral, and the engine shall be off. (CAUTION: If the transfer arms are raised more than 15 degrees, the loader must be engaged with the pallet or launcher to prevent overturn.)

4.5.3.7.3 Superstructure operating speeds. To determine conformance to 3.3.8.3, the loader shall be loaded with 3 missiles on a relatively smooth, level, firm surface, and the control levers shall be placed at their maximum settings. The loader shall then perform as specified in table IV at the specified engine speeds.

4.5.3.8 Loading operations.

4.5.3.8.1 Missile handling. To determine conformance to 3.3.9.1, the loader shall load and unload missiles under the following conditions:

a. With the loader and launcher positioned on a slope of 10 degrees, the loader shall index to the launcher and load or unload the launcher with 1, 2, or 3 missiles. When loading or unloading less than 3 missiles, the missiles may be guided by hand if necessary.
b. With the pallet mounted on an XM-390 type trailer, and with the loader and trailer positioned on a slope of 10 degrees, the loader shall index to the pallet and load and unload the pallet with 1, 2, or 3 missiles. When loading or unloading less than 3 missiles, the missiles may be guided by hand if necessary.
c. With the loader and pallet positioned on the ground, on a slope of 10 degrees, the loader shall be capable of indexing to the pallet and loading and unloading the pallet with 1, 2, or 3 missiles. When loading or unloading less than 3 missiles, the missiles may be guided by hand if necessary.

NOTE: The trailer mounted pallet must be approximately leveled when the trailer pintle is down slope or when located on a slide slope. When the trailer pintle is up slope, the trailer bed must be approximately parallel to the ground.

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4.5.3.8.2 Pallet loading. To determine conformance to 3.3.9.2, the loader, with the suspension locked, shall engage and transfer an empty pallet from an XM-390 trailer to the ground and vice versa.

4.5.3.8.3 Proof load. To determine conformance to 3.3.9.3, the loader shall be subjected to the following proof load tests:

a. Suspension system shall be locked out prior to proof load testing and will remain in that position until all proof load testing is completed. Attach a 1000 pound (450 kg) ballast to the loader pintle hook prior to proof loading.
b. Increase the weight of each test missile to a minimum of 1526 pounds (692 kg) and a maximum of 1580 pounds (717 kg). The weights shall be added to maintain the missile center-of-gravity (c.g.).
c. Index the loader to the pallet, loaded with three missiles with weights, and latch hoisting beam to missile. Install two (2) safety straps around hoisting beam and missiles. The straps shall be fastened to prevent the missiles from falling in event of failure, but shall not carry the missile weight during proof load testing.

-WARNING-

NOTE: Caution must be exercised by the operator during proof load testing to meter the control valves to prevent abrupt superstructure motions. Engine speed shall be 700 to 800 rpm.
CAUTION: Personnel not directly engaged in the proof loading exercises are not to enter an area of 30 feet (9 m) in any direction of the loader during proof load testing.
CAUTION: All proof load testing exercises will be conducted with the driver's protective canopy in place and with the operator under that canopy.

d. With the loader indexed to the pallet on the ground, retract the extension cylinder fully. Extend the extension cylinder until missiles seat on the pallet.
e. Repeat "d" above for two (2) cycles.
f. With the loader indexed to the pallet on the ground, retract the extension cylinder fully and then break index by elevating the boom. Place missiles in the transport position.
CAUTION: Exercise caution to keep missile load as far aft on loader as possible by using elevation and extension controls in sequence.
g. Raise missiles enough to allow azimuth rotation of superstructure at 0 degrees. Move missiles to full azimuth right. Move missiles to azimuth left.
h. Repeat "g" above for two (2) cycles and then center missiles in azimuth.

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i. Index loader to pallet, seat missiles on pallet, release latch on "C" missile only, and remove straps from "C" missile. Return superstructure with "A" and "B" missiles to transport position.
j. Position loader under a test fixture or crane, and install sling to fixture or crane and loader superstructure yoke. Sling shall have enough slack to allow superstructure to roll fully right, but hold superstructure in the event of failure.
k. With missiles elevated sufficiently to permit roll motions, roll missiles full right.
l. Repeat "k" above for two (2) cycles and then return to center.
m. Remove sling. Index loader to pallet and transfer missiles onto pallet. Remove straps, latch missiles to pallet, release loader latches, and retract loader superstructure.
n. Recheck loader alignment on pallet/alignment fixture.

4.5.3.9 Crane operation. With the suspension locked, the loader, when fitted with crane accessories and adjusted for crane operation, shall function as a crane on slopes up to 10 degrees. Nonconformance to 3.3.10 shall constitute failure of the test.

4.5.3.9.1 Hoisting cable. To determine conformance to 3.3.10.1, the extension control lever shall be moved forward and backward to respectively raise and lower a maximum load of 1580 lbs (717 kg).

4.5.3.9.2 Boom swing. To determine conformance to 3.3.10.2, the maximum load of 1580 lbs shall be applied to the boom, and the azimuth control lever shall be turned clockwise and counterclockwise.

4.5.3.9.3 Boom elevation. To determine conformance to 3.3.10.3, the maximum load of 1580 lbs shall be applied to the boom, and the elevation control lever shall be moved forward and backward.

4.5.3.9.4 Hoisting hook position. With the crane rigged without cable stay and load, the crane boom (ramp beam) shall first be fully depressed (elevation cylinder fully extended) with the transfer arm sockets seated on the index boom trunnion (hoisting cable is fully extended). The distance from the lowest point of the hoisting hook to the ground shall then be measured. Next, the crane boom shall be fully elevated (index boom resting on the rear missile support) with the extension cylinder fully extended (hoisting cable is fully retracted). The distance from the lowest point of the hoisting hook to the ground shall then be measured. Nonconformance to 3.3.10.4 shall constitute failure of the test.

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4.5.3.9.5 Rated capacity. To determine conformance to 3.3.10.5, the loader crane, in a non-mobile condition on a slope of 10 degrees, shall lift a minimum load of 2500 lbs (1130 kg). The loader shall then be operated at reduced speed on the 10 degree slope while lifting a load of 1600 lbs (730 kg). (CAUTION: Use the crane sling to restrain the load and do not lift the load more than one foot above the ground.)

4.5.3.10 Auxiliary equipment. The auxiliary equipment specified in 6.3, shall be used to check the compatibility of the loader. Nonconformance to 3.3.11 shall constitute failure of the test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The loader-transporter is intended to transfer HAWK guided missiles from pallets to launcher and vice versa, and to otherwise handle missiles and pallets as may be necessary within a missile battery. It is also intended for limited use as a crane and for short distance towing.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. If first article is required (see 3.1).
d. If OVM is required (see 3.2.6).
e. Type of surface treatment if other than as specified (see 3.2.7).
f. Color required (see 3.2.7).
g. If CARC paint is required (see 3.2.7).
h. Identification markings if other than as specified (see 3.2.8).

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i. If preproduction inspection is required (see 4.2.1).
j. If initial production inspection is required (see 4.2.2).
k. If pretest run is required (see 4.2.3).
l. Packaging requirements (see 5.1).

6.3 Auxiliary equipment. The auxiliary equipment and the related drawings are as follows:

Auxiliary equipment (see 4.5.3.10)	Drawing
Guided Missile, air Defense: XM3E1 Standard Type	9071755
Launcher, Zero Length, Guided Missile:	9098880
Pallet, Loading and Storage, Guided Missile:	9196208
Truck, Cargo, 2-1/2 Ton, M36	8750400
Chassis, Trailer, 2 Ton, M390C	8736315
Ramp	9098330
Crane	9098300
Adapter, hoist	10916330

6.4 Definitions. For the purpose of this specification, the following definitions will apply:

a. Normal loader operation. Normal loader operation will refer to operation associated with the loaders primary tactical function within a missile battery, namely that of transferring 1, 2, and 3 missiles at a time from storage pallets to launchers and vice versa, and transferring pallets from ground to XM390 trailers and the reverse. For such operation, indexing forks will point toward front of loader, superstructure will respond to hydraulic controls as specified in 3.2.4.2 and superstructure electrical circuits will operate in accordance with section 3.2.5. Normal operation will involve indexing, positioning of superstructure on pallets or launchers for loading and unloading of missiles, latching of missiles to superstructure and release therefrom, transport of missiles latched to superstructure, and latching to, transporting and unloading empty pallets on XM390 trailers or the reverse.
b. Normal loader transport position. Normal loader transport position will refer to the position of the superstructure during missile transport by the loader (in normal loader operation) wherein the latched missile(s) are held firmly in loader saddles with full weight of each missile carried by the applicable front and rear loader missile supports.

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c. Final extension phase. Final extension phase will refer to operation of superstructure in extension and retraction (in normal loader operation) during period when reduced extension pressure feature is operative as specified in 3.2.4.4. In missile transfer operations, loader is indexed to launcher or pallet during the final phase of extension.

d. Crane operation. Crane operation will refer to operation of the loader when loader is rigged with items as specified (including ramp), and adjusted for crane operation. The loader will be adapted as follows for crane operation:

1. Place forward and rear center missile supports in stowed positions (see 3.2.4.7 and 3.2.4.8).
2. Lockout carrier suspension system (see 3.2.4.1.1).
3. Unpin azimuth cylinder rod boom support, rotate boom through 3200 mils and repin to azimuth cylinder.
4. Place roll lockout in “crane” position (see 3.2.4.1.2).
5. Extend superstructure until transfer arms contact link assembly and pin together using links provided.
6. Assemble crane attachments to ramp. Place ends of ramp in crane support bracket and pin.
7. Place crane support bracket on hoisting support boom trunnion shaft. Using quick release pins, pin front stays to boom support assembly and pin fairlead sheave assembly to center hoisting beam housing.
8. Using loader power, lift ramp with crane cable and pin rear cable stays to loader index forks.

6.5 Subject term (key word) listing.

Anti-aircraft
Azimuth
Boom
Carrier
Crane
Superstructure
Towing

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-46821C(AT)	2. DOCUMENT DATE (YYMMDD) 960920

3. DOCUMENT TITLE
Loader-Transporter, Guided Missile: XM501E3 (Improved Hawk)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial: (810) 574-8745 (2) AUTOVON: 786-8745
c. ADDRESS (Include Zip Code) Commander U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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